DSA 210: TERM PROJECET

Phase 4: Applying Machine Learning Methods on The Data Set

1.Introduction

In this phase of the project, several supervised machine learning methods are applied to analyze and model the behavior of Market Clearing Price (MCP) and System Marginal Price (SMP) in the dataset. The methods considered include:

1. Linear Regression
2. Logistic Regression
3. K Nearest Neighbors (KNN)
4. Decision Trees

Before sharing the conclusions of these algorithms, an important distinction should be clarified on the nature of the SMP and MCP. MCP is a value that is determined by EPIAS, for each hour of each day and it is independent of the production behavior. On the other hand, SMPs occur as a penalty cost in the system for either overproduction or underproduction.

It should also be noted that these prices are based on Turkish Liras, whereas electricity prices are driven by the global factors and are seriously affected by inflation.

Both mentioned aspects of this data set clearly affect the outcome of the ML processes.

2.Linear Regression

Linear regression is applied as a baseline model to analyze the relationships between SMP, MCP, and hour and month features. Three different model configurations are considered:

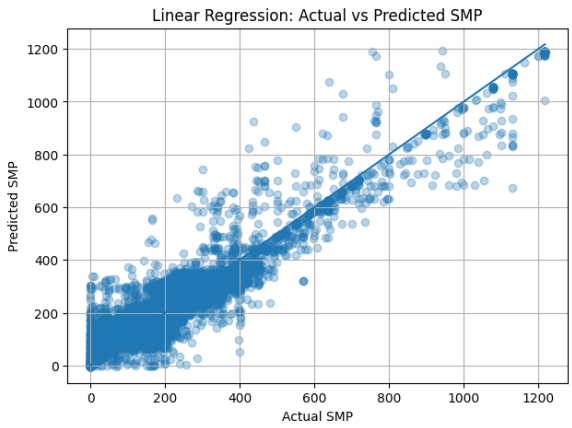
- SMP is predicted using MCP, hour, and month as input features
- SMP is predicted using only hour and month as input features
- MCP is predicted using hour and month as input features

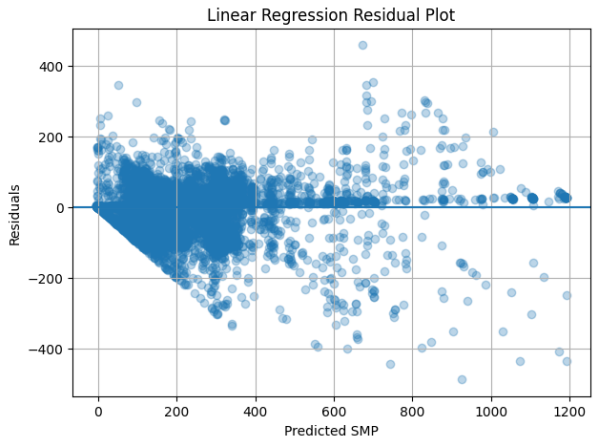
A. SMP Regression with MCP as a Feature

The linear regression model using MCP, hour, and month achieves strong predictive performance, with an R^2 value of approximately 0.87. This indicates that the model can explain a large portion of the variability in SMP.

The results of this analysis were as the following:

```
Linear Regression: SMP using MCP, Hour, Month
MAE: 36.280476039763975
R2: 0.8724330984454826
```





It should be noted that the dispersions that occur mostly while SMP Values are high.The residual plot further supports this observation, showing increasing variance for higher predicted values. This indicates that while the model captures the general relationship effectively, it struggles with extreme price levels.

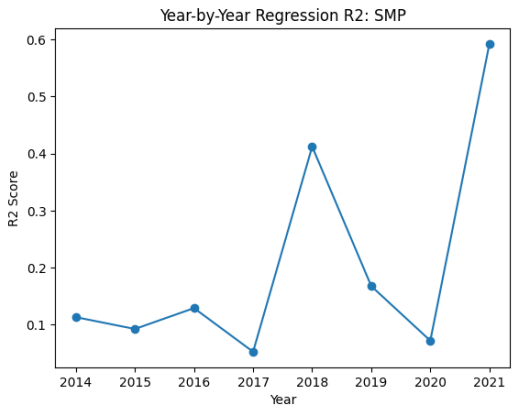
B. SMP Regression

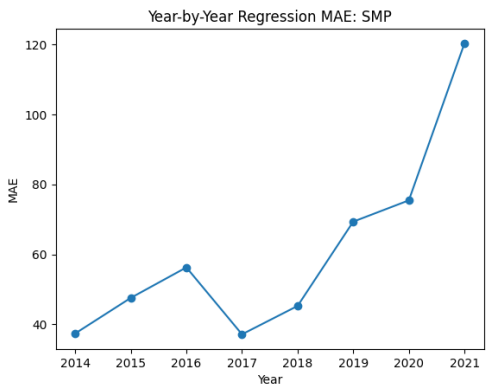
The regression results clearly show that SMP cannot be explained using only hour and month features. However, when MCP is included as an input variable, the model achieves a significantly higher performance.

```
SMP Regression
MAE: 102.87463759609442
R2: 0.07084354039035523
```

To support this conclusion and to avoid the inflation effect, the year by year analysis was also conducted and the same result was obtained.

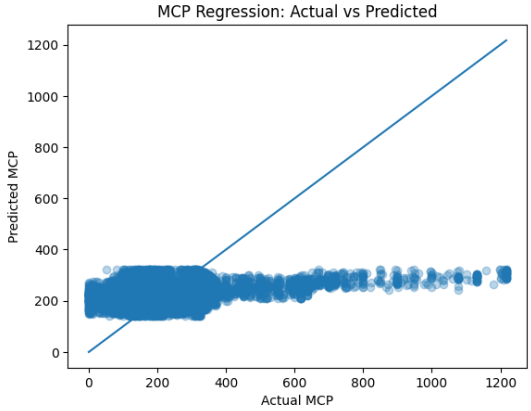
	Year	R2	MAE
0	2014	0.113023	37.349043
1	2015	0.092370	47.533035
2	2016	0.128964	56.248326
3	2017	0.052363	37.115186
4	2018	0.412237	45.236218
5	2019	0.167570	69.333495
6	2020	0.071816	75.423333
7	2021	0.592445	120.317701



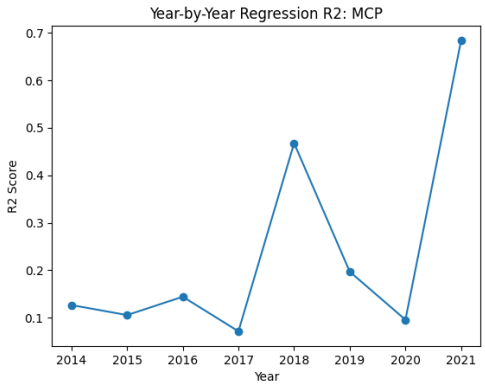


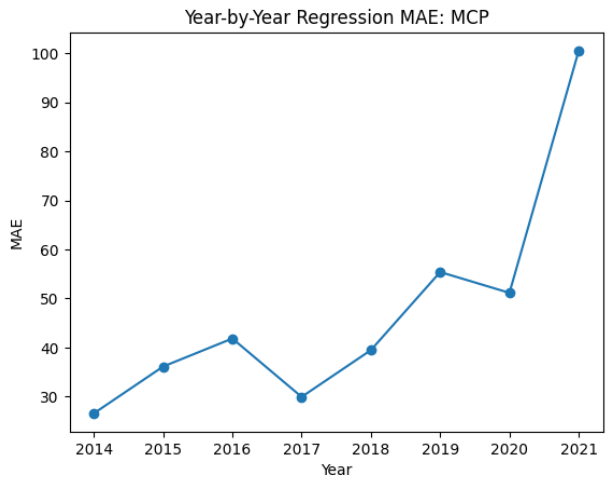
C. MCP Regression Analysis

Above conclusions for the SMP can also be made for the SMP since they both follow the same structure when it comes to deviations within themselves. The results can be supported by the below graphs.



	Year	R2	MAE
0	2014	0.126910	26.598315
1	2015	0.105798	36.140498
2	2016	0.144386	41.851073
3	2017	0.071630	29.956221
4	2018	0.467377	39.487403
5	2019	0.196970	55.399422
6	2020	0.096025	51.155419
7	2021	0.683707	100.475214





3.Logistic Regression

Logistic regression was not suitable for this dataset because SMP and MCP values are continuous time series data points, whereas logistic regression is designed for classification problems with discrete outputs.

Even though the accuracy level seemed considerable, this occurred because the model predicted the median value of 0.

```
Median threshold: 0.0
High_Deviation
0    41331
1    28605
Name: count, dtype: int64
Logistic Regression Results
Accuracy: 0.5899342293394338
Confusion Matrix:
[[8192   75]
 [5661   60]]
Classification Report:
              precision    recall  f1-score   support

           0       0.59      0.99      0.74      8267
           1       0.44      0.01      0.02      5721

    accuracy                           0.59     13988
   macro avg       0.52      0.50      0.38     13988
weighted avg       0.53      0.59      0.45     13988
```

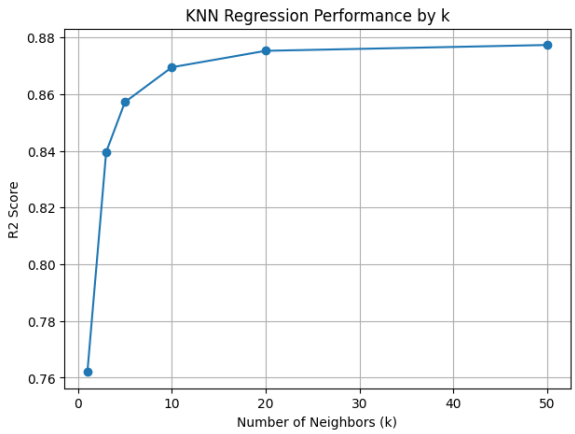
4. K Nearest Neighbor Algorithm

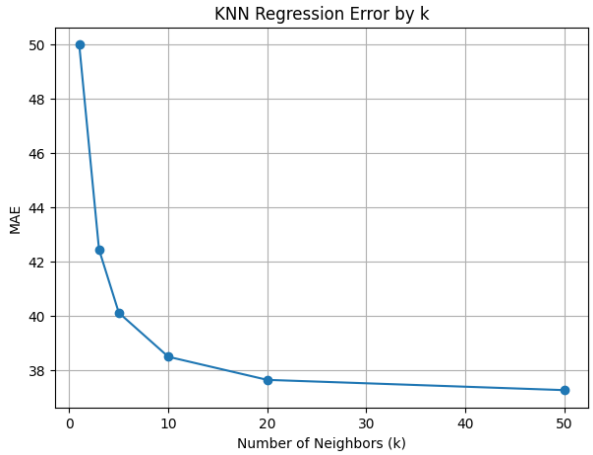
As mentioned before, the data is biased towards price fluctuations. Therefore, algorithms such as kNN methods and decision trees can do better predictions. The algorithm was applied several times, with different neighborhood sizes.

The below graph and table shows the accuracy of 6 iterations that was completed with different k values.

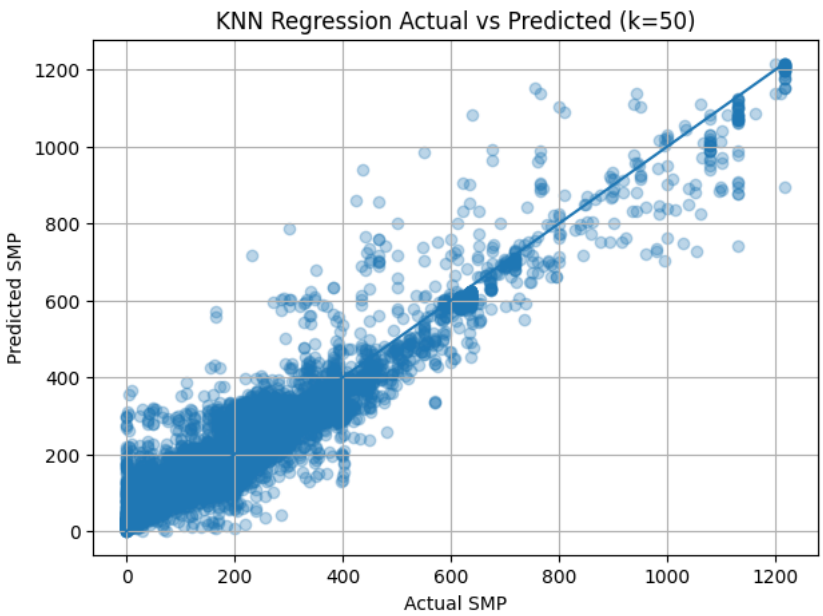
```
   k         MAE         R2
0   1  49.996400  0.762041
1   3  42.449944  0.839581
2   5  40.119241  0.857182
3  10  38.498735  0.869465
4  20  37.647595  0.875251
5  50  37.270047  0.877313
Best k: 50
```

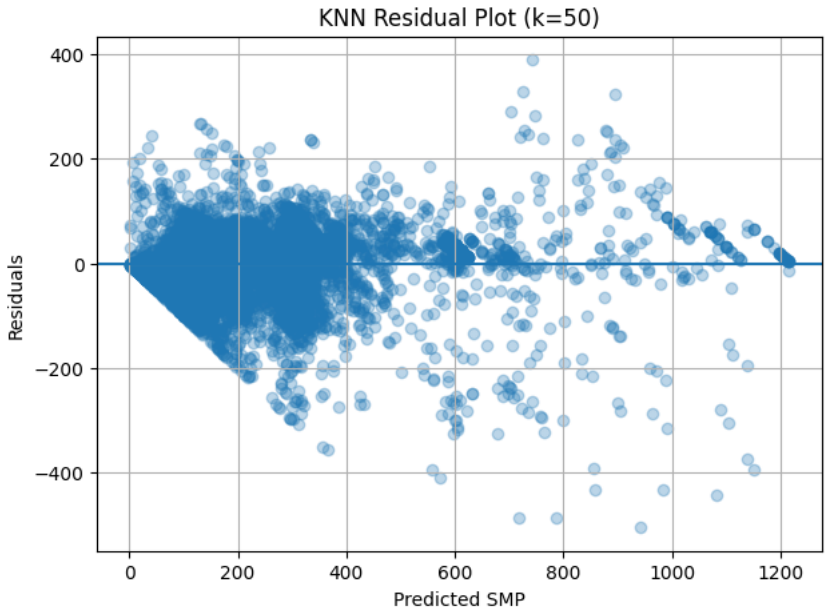
```
Best KNN Model
MAE: 37.270046983128395
R2: 0.8773132070541497
```





Accuracy of the predictions can be observed with the below graphs.



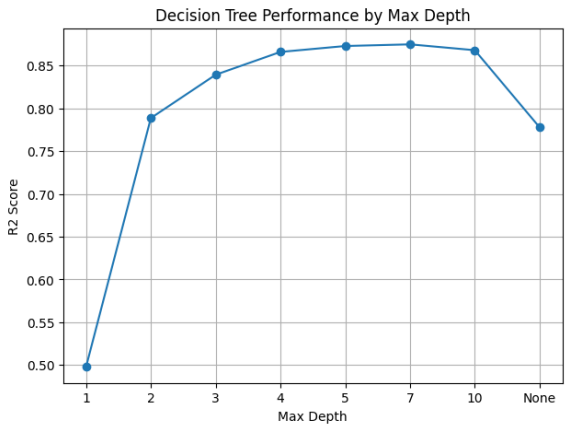


5.Decision Tree Algorithm

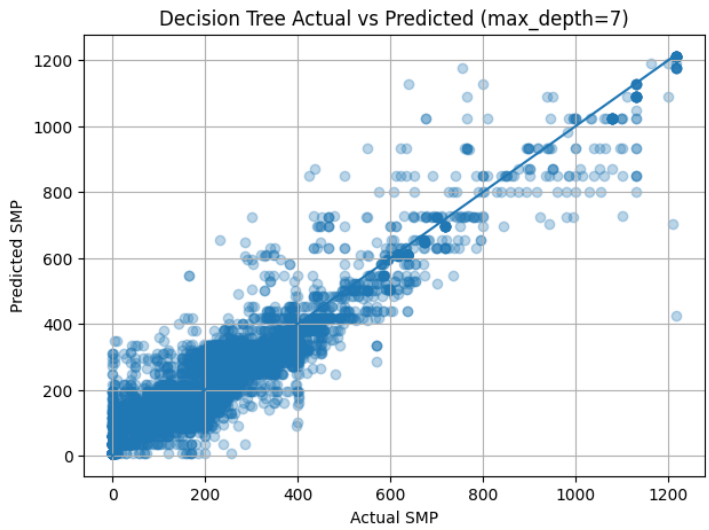
Decision tree algorithm is used to capture potential nonlinear relationships and to identify the importance of input features. The algorithm was applied through several iterations, while in each iteration the maximum depth size changed.

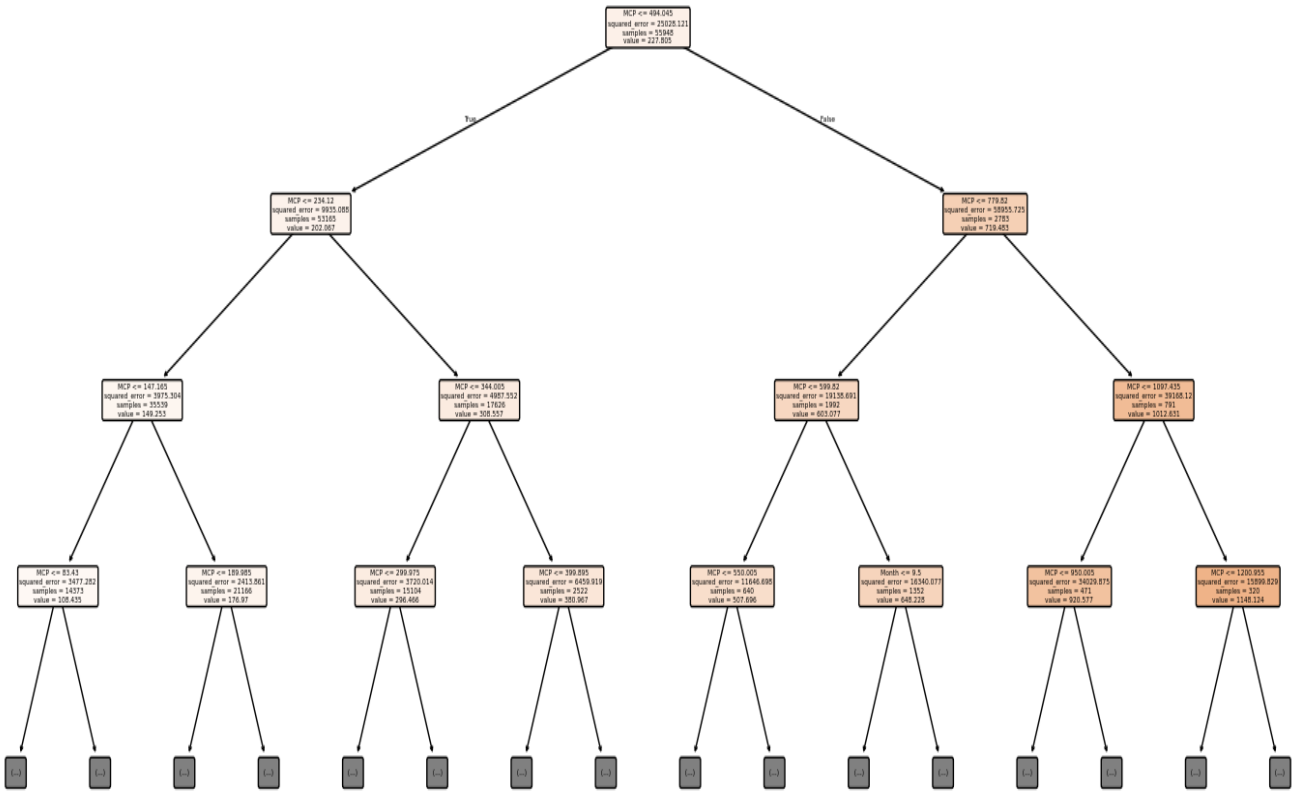
```
   max_depth        MAE        R2
0          1  84.240702  0.497833
1          2  53.357532  0.788763
2          3  45.824895  0.839401
3          4  39.682784  0.866083
4          5  37.953204  0.872993
5          7  37.248297  0.874947
6         10  37.538428  0.868088
7       None  48.594820  0.777872
Best max_depth: 7

Best Decision Tree Model
MAE: 37.248296502977574
R2: 0.8749471333360207
```



The modeled behaved more accurately when maximum depth was 7. A decision tree with this parameter was calculated and the below results were obtained.



For better visualization, the decision tree attached can be used as a reference.

Decision Tree Structure - First 3 Levels